E-Commerce Customer Purchase Analysis Project

Submitted By:

Name: Abhinav Sharma

Roll Number: 2305756

Batch/Program: B.Tech (CSE)

Submitted To: Sachin Sir

Date: 20 April 2026

Objective / Problem Statement

The marketing department of an online shopping platform wants to analyze customer purchasing behavior to improve promotional campaigns and identify valuable customers.

The platform records information about customer purchases including customer details, product purchased, quantity, price, and city location.

This project aims to answer the following key business questions:

- Which customers spend the most money?
- Which products are most popular?
- Which cities generate the most revenue?

The analysis will help the marketing team make data-driven decisions for better targeting and increasing sales.

Solution / Features

Tech Stack Used:

- Oracle SQL (for database creation and queries)
- Microsoft Excel (for data cleaning and Pivot Table analysis)
- Python (Pandas for data analysis and Matplotlib for visualizations)

Key Features Implemented:

- Calculated Total Purchase amount for each transaction
- Identified top spending customers
- Analyzed product popularity based on quantity and revenue
- Performed city-wise revenue analysis
- Created meaningful visualizations (Bar chart and Pie chart)
- Provided actionable business insights and marketing recommendations

This project demonstrates end-to-end data analysis skills covering data extraction, transformation, analysis, and visualization.

Methodology / Implementation

1. SQL Implementation (Oracle) A table named purchases was created in Oracle Database. Sample data was inserted and the following queries were executed:

- Display all purchase records
- Total spending per customer
- Most purchased products
- City-wise revenue analysis

2. Excel Implementation

- Loaded the dataset into Excel
- Created a new column total_purchase using formula (Quantity × Price)
- Built Pivot Tables for:
    - Product-wise sales summary
    - City-wise revenue summary
- Added charts for better visualization

3. Python Implementation

- Loaded the CSV file using Pandas
- Created total_purchase column
- Performed grouping and aggregation to find top customers, popular products, and city revenue
- Generated two visualizations:
    - Bar chart for Product Sales by Revenue
    - Pie chart for City-wise Revenue Distribution

Insights & Recommendations

Key Findings:

- Sneha from Mumbai is the highest spending customer (₹60,000).
- Mobile is the most popular product in terms of both quantity and revenue.
- Mumbai generates the highest revenue among all cities.
- Pune is the second highest revenue-generating city.

Marketing Recommendations:

- Focus maximum promotional budget on Mumbai and Pune cities.
- Target high-value customers (like Sneha) with loyalty programs and special discounts.
- Create bundle offers combining Mobile with Headphones or other accessories.
- Run city-specific campaigns in high-revenue areas for better ROI.
- Promote Mobile phones aggressively as it is the top-selling product.

These insights can help the marketing team increase customer retention and boost overall sales.

Screenshots

```
SELECT
    city,
    SUM(quantity * price) AS total_revenue,
    COUNT(*) AS total_orders
FROM purchases
GROUP BY city
ORDER BY total_revenue DESC
```

CITY	TOTAL_REVENUE	TOTAL_ORDERS
Mumbai	100000	2
Nagpur	25000	1
Pune	24000	2

3 rows selected. 0.01 seconds

```
SELECT
    product,
    SUM(quantity) AS total_quantity_sold
FROM purchases
GROUP BY product
ORDER BY total_quantity_sold DESC
```

PRODUCT	TOTAL_QUANTITY_SOLD
Mobile	3
Headphones	2
Tablet	1
Laptop	1

4 rows selected. 0.01 seconds

```
SELECT
    customer_id,
    customer_name,
    city,
    SUM(quantity * price) AS total_spending
FROM purchases
GROUP BY customer_id, customer_name, city
ORDER BY total_spending DESC
```

CUSTOMER_ID	CUSTOMER_NAME	CITY	TOTAL_SPENDING
2	Sneha	Mumbai	60000
5	Meena	Mumbai	40000
4	Ritu	Nagpur	25000
1	Arjun	Pune	20000
3	Akash	Pune	4000

5 rows selected. 0.01 seconds

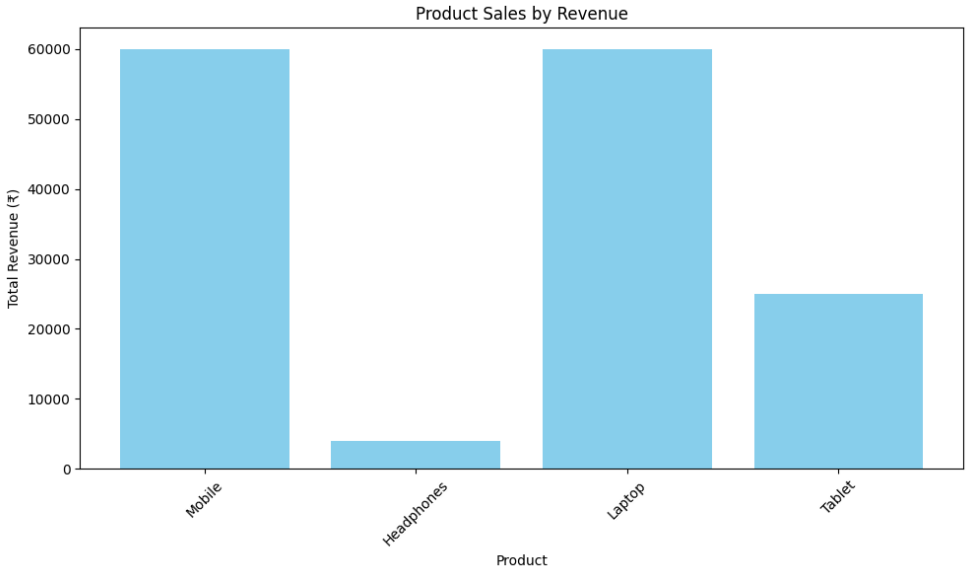
```
SELECT * FROM purchases
```

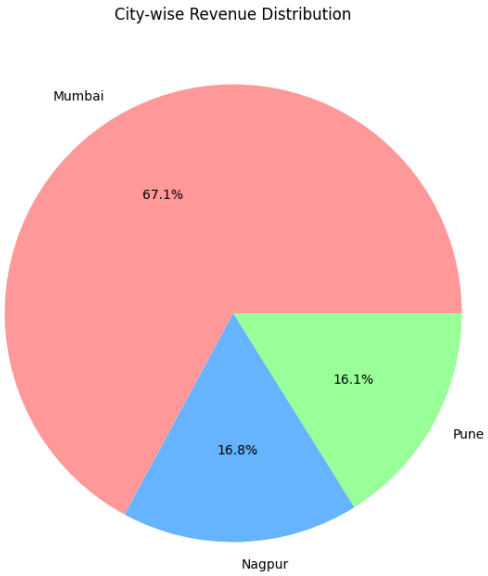
CUSTOMER_ID	CUSTOMER_NAME	CITY	PRODUCT	QUANTITY	PRICE
1	Arjun	Pune	Mobile	1	20000
2	Sneha	Mumbai	Laptop	1	60000
3	Akash	Pune	Headphones	2	2000
4	Ritu	Nagpur	Tablet	1	25000
5	Meena	Mumbai	Mobile	2	20000

	A	B	C	D	E	F	G
1	customer_id	customer_name	city	product	quantity	price	total_purchase
2	1	Arjun	Pune	Mobile	1	20000	20000
3	2	Sneha	Mumbai	Laptop	1	60000	60000
4	3	Akash	Pune	Headphones	2	2000	4000
5	4	Ritu	Nagpur	Tablet	1	25000	25000
6	5	Meena	Mumbai	Mobile	2	20000	40000

city	Sum of total_purchase	Sum of customer_id
Mumbai	100000	7
Nagpur	25000	4
Pune	24000	4
Grand Total	**149000**	**15**

product	Sum of total_purchase	Sum of quantity
Headphones	4000	2
Laptop	60000	1
Mobile	60000	3
Tablet	25000	1
Grand Total	**149000**	**7**





Conclusion & Future Improvements

Conclusion: This project successfully demonstrated real-world E-Commerce Customer Purchase Analysis using Oracle SQL, Excel, and Python. It helped in understanding how data analysis can directly support marketing decisions and improve business performance.

Future Improvements:

- Implement Customer Segmentation using K-Means clustering
- Build an interactive dashboard using Power BI or Tableau
- Develop a Recommendation System for products
- Add predictive analysis (e.g., forecasting future sales)
- Deploy the project as a web application using Streamlit or Flask

This project has built a strong foundation in Data Fundamentals and will be very useful for future data-related roles.